
POWER IN (5V DC terminal or USB, ORed Schottky diodes)
USB-C (native USB flashing + ESD)
THERMOCOUPLE 1 (control) MAX31856 (T− floats to J3, biased via BIAS)
THERMOCOUPLE 2 (load) MAX31856 (T− floats to J8, biased via BIAS)
ESP32-S3-WROOM-1 (GPIOs = firmware Kconfig defaults)
SSR DRIVE x2 (low-side AO3400A; indicator LED across each terminal)
CT CURRENT SENSING ADE7953, I2C, current-only (no mains − VP/VN to DNP J13)
WS2812B STATUS LED (VDD dropped ~4.6V for 3.3V data margin)
ALARM BUZZER
HEADERS − DISPLAY+TOUCH(J5, 14-pin) NAV(J6) AUX+I2C(J7)
PROTECTED INPUTS x3 lid/gas-flow/spare (1k + 10k pull-up + 100nF each, SRV05-4 TVS)
MOUNTING / POWER FLAGS
RESET / BOOT / DECOUPLING
Bisque kiln controller − ESP32-S3-WROOM-1U-N16R2 (rev B)
SSR terminals J4/J9: pin1 = SSR_EN (board +5V, watchdog-gated), pin2 = the switched low side. Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance. NOT opto-isolated − rev B tried that and reverted it: an opto only isolates if the SSR loop is powered off-board, and this board powers it.
TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K− − both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required: two grounded-junction probes in one kiln would loop through it.
J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND − dry contact each channel to GND
Display J5 (14-pin, ST7796S + XPT2046 touch module − LCDWIKI MSP4021):
1=+5V 2=GND 3=CS 4=RST 5=DC 6=SDI/MOSI 7=SCK 8=BL 9=SDO/MISO
10=T_CLK 11=T_CS 12=T_DIN 13=T_DO 14=T_IRQ. PIN 1 IS +5V, NOT 3V3 − the module regulates on board; do not wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31856s (R39−R43 damping)
J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N − one CT clamp per SSR zone.
J13: DNP SELV voltage sense header. NOT mains-rated, not fitted − do not wire to AC mains. Y1/C25/C26 load caps are an ASSUMED value (unverified C_L); see design.py comment. ADE7953 I2C address 0x38 collides with PCF8574A.
AUX OUTPUT BANK (ULN2003 vent/purge/spare; COM−>AUX_VP, SJ1 links to +5V)
TOUCH DAMPING + I2C EXPANSION (Task 11) R39−43 damp the shared SPI2\nbus for the display module's XPT2046 (not on this board); J14 Qwiic +\nJ7 5−8 (0.1") share the I2C bus, pulled up by R44/R45
HARDWARE WATCHDOG C38/D7/C39/R46 diode charge pump on WDT_KICK holds\nQ3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies\nSSR_EN − the +5V rail feeding BOTH SSR terminals and both indicators.\nR47 is the fail-safe pull-up. SJ2 = bring-up defeat, REMOVE for service.
4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project
Sheet: /
File: bisque-controller.kicad_sch
Title: Bisque Kiln Controller
Size: A1
Date: 2026-07-20
Rev: B
KiCad E.D.A. 10.0.5
Id: 1/1